Objectives of Ch 6 – Normal Distribution / Continuous Random Variable

- To compute probabilities from the normal distribution.
- How to use the normal distribution to solve business problems.

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Recap of Ch 5

BINOMIAL: Each trial results in one of two possible outcomes (success or failure)

POISSON: Each trial has constant probability of event of interest (success)

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Recap of Ch 5-Examples

Binomial:

A pharmaceutical lab states that a drug causes negative side effects in 3 of every 100 patients. To confirm this affirmation, another laboratory chooses 5 people at random who have consumed the drug. What is the probability of the following events?

None of the five patients experience side effects.

At least two experience side effects.

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Recap of Ch 5

Poisson:

An insurance company was reviewing its rates to cover race car drivers. Over a 20-year period, an average of 0.42 drivers were involved in serious accidents per year. Based on this information, complete the table below:

Number of injuries	Probability
0	
2	
Less than 3	

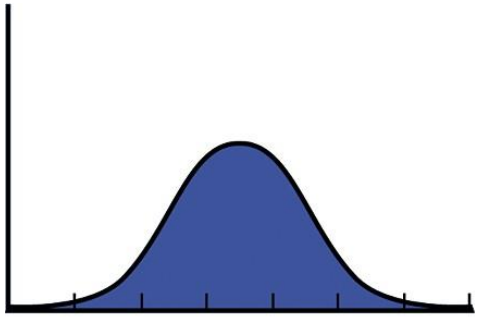
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Continuous Probability Distributions

- A continuous variable is a variable that can assume any value on a continuum (can assume an uncountable number of values):
  - thickness of an item.
  - time required to complete a task.
  - temperature of a solution.
  - height, in inches.

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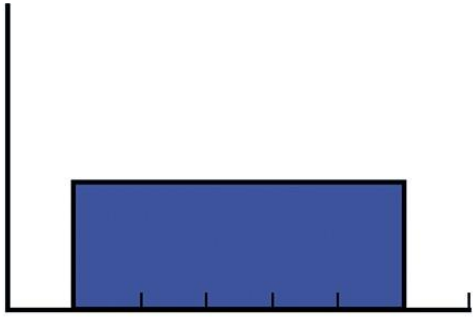
Continuous Probability Distributions Vary By Shape



Values of X

Normal Distribution

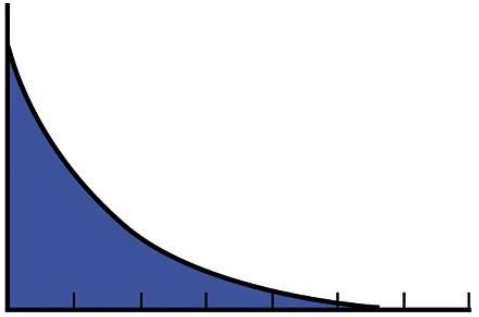
- Symmetrical
- Bell-shaped
- Ranges from negative to positive infinity



Values of X

Uniform Distribution

- Symmetrical
- Also known as Rectangular Distribution
- Every value between the smallest & largest is equally likely



Values of X

Exponential Distribution

- Right skewed
- Mean > Median
- Ranges from zero to positive infinity

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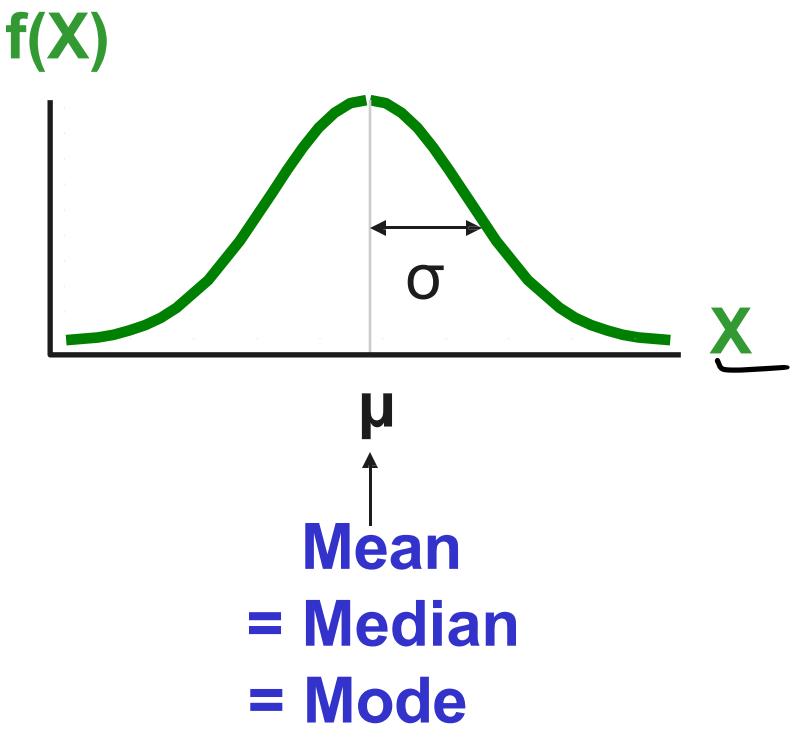
The Normal Distribution

- **Bell Shaped.**
- **Symmetrical. .**
- **Mean, Median and Mode are Equal.**

Location is determined by the mean, μ.

Spread is determined by the standard deviation, σ.

The random variable has an infinite theoretical range: -∞ to +∞.



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The Normal Distribution Density Function

- The formula for the normal probability density function is:

$$f(X) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{(X-\mu)}{\sigma}\right)^2}$$

Where e = the mathematical constant approximated by 2.71828

π = the mathematical constant approximated by 3.14159

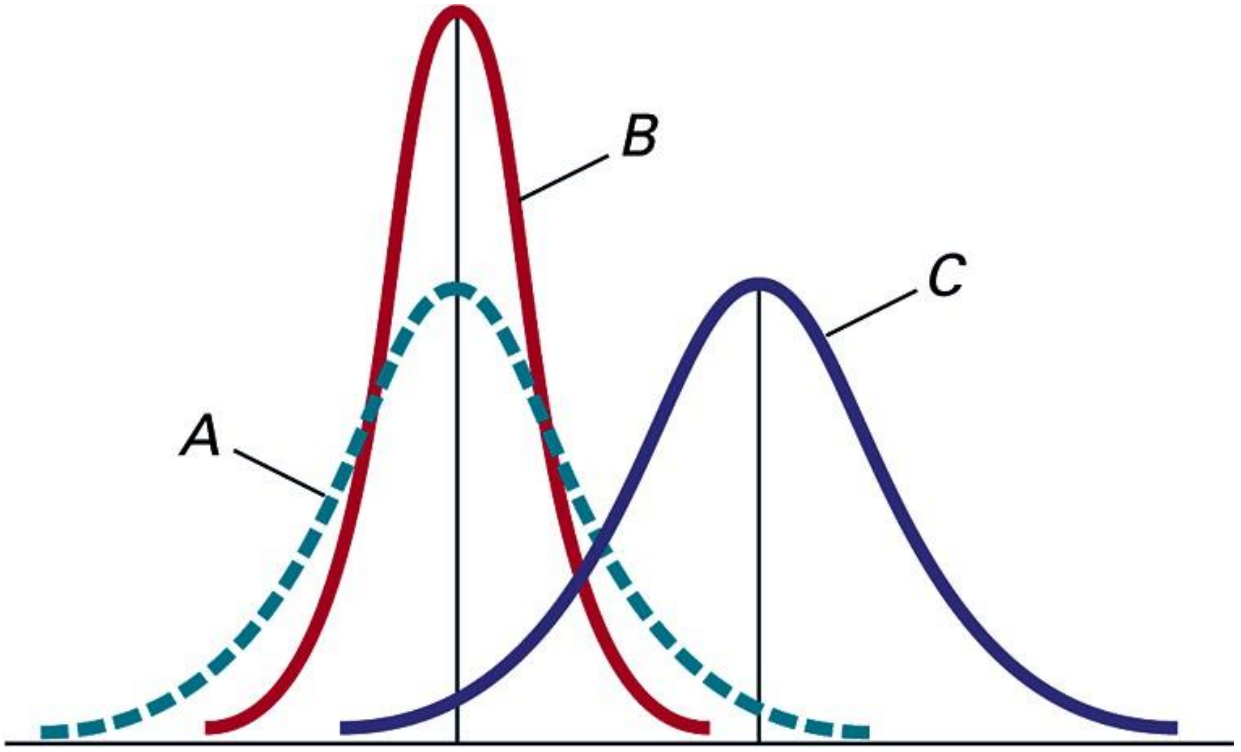
μ = the population mean

σ = the population standard deviation

X = any value of the continuous variable

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By varying the parameters μ and σ, we obtain different normal distributions

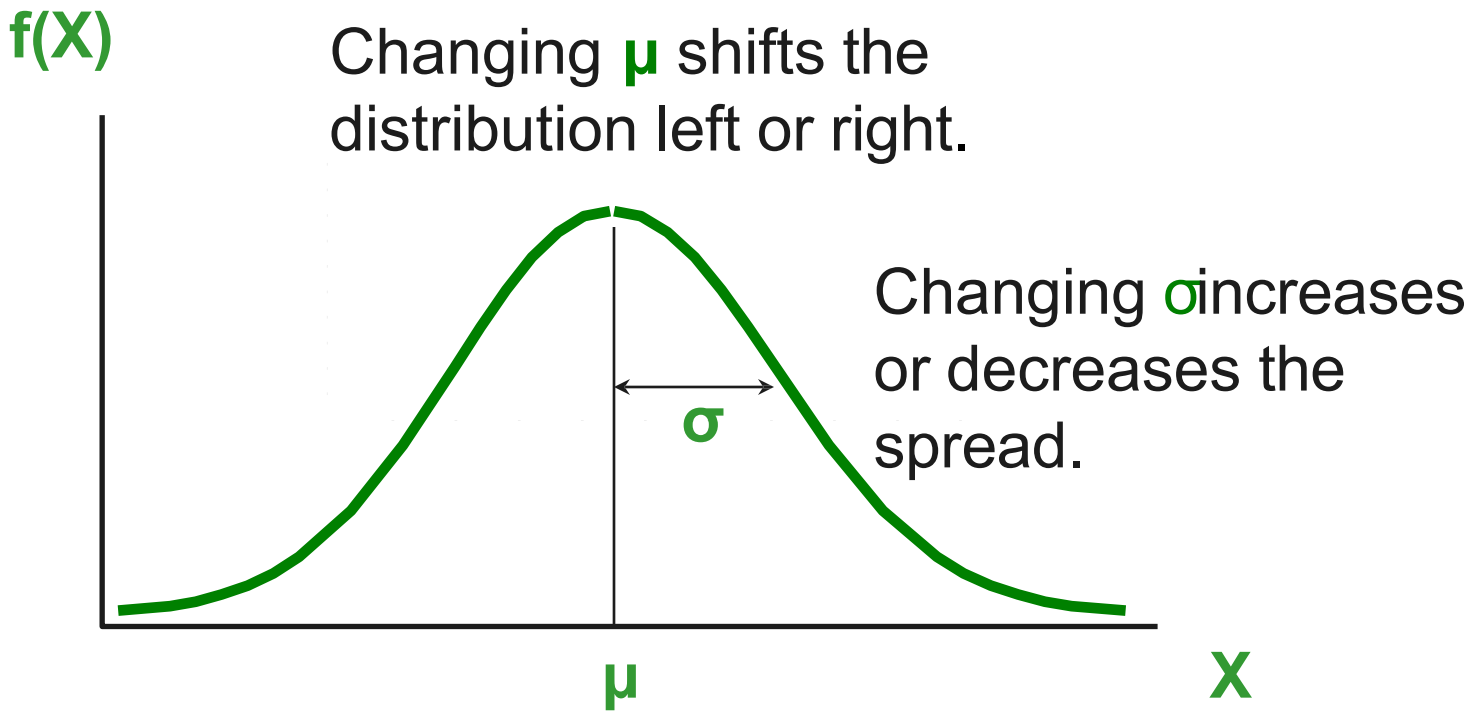


A and B have the same mean but different standard deviations.

B and C have different means and different standard deviations.

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The Normal Distribution Shape



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The Standardized Normal

- Any normal distribution (with any mean and standard deviation combination) can be transformed into the standardized normal distribution (Z).

- To compute normal probabilities need to transform X units into Z units.

- The standardized normal distribution (Z) has a mean of 0 and a standard deviation of 1.

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Translation to the Standardized Normal Distribution

- Translate from X to the standardized normal (the "Z" distribution) by subtracting the mean of X and dividing by its standard deviation:

$$Z = \frac{X - \mu}{\sigma}$$

The Z distribution always has mean = 0 and standard deviation = 1.

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The Standardized Normal Probability Density Function

- The formula for the standardized normal probability density function is:

$$f(Z) = \frac{1}{\sqrt{2\pi}} e^{-(1/2)Z^2}$$

Where e = the mathematical constant approximated by 2.71828

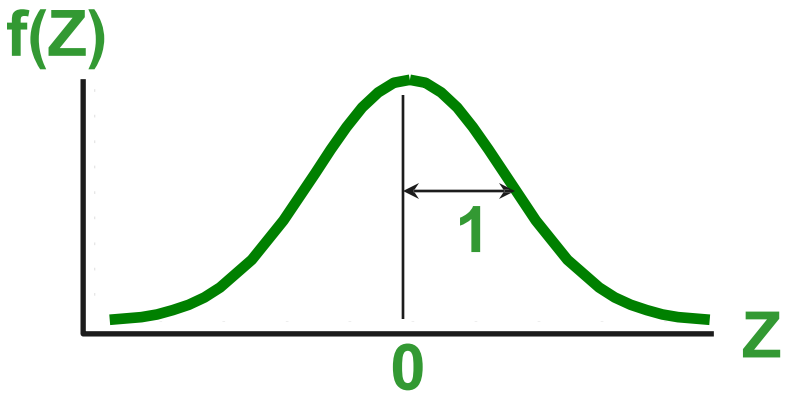
π = the mathematical constant approximated by 3.14159

Z = any value of the standardized normal distribution

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The Standardized Normal Distribution

- Also known as the “Z” distribution.
- Mean is 0.
- Standard Deviation is 1.



Values above the mean have positive Z-values.
Values below the mean have negative Z-values.

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Example

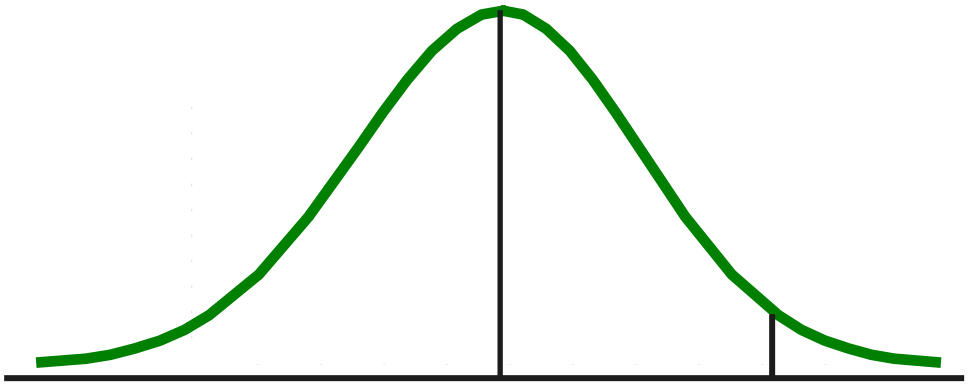
- If X is distributed normally with mean of $100 and standard deviation of $50, the Z value for X = $200 is:

$$Z = \frac{X - \mu}{\sigma} = \frac{\$200 - \$100}{\$50} = 2.0$$

- This says that X = $200 is two standard deviations (2 increments of $50 units) above the mean of $100.

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Comparing X and Z units



$100 $200 $X (μ = $100, σ = $50)

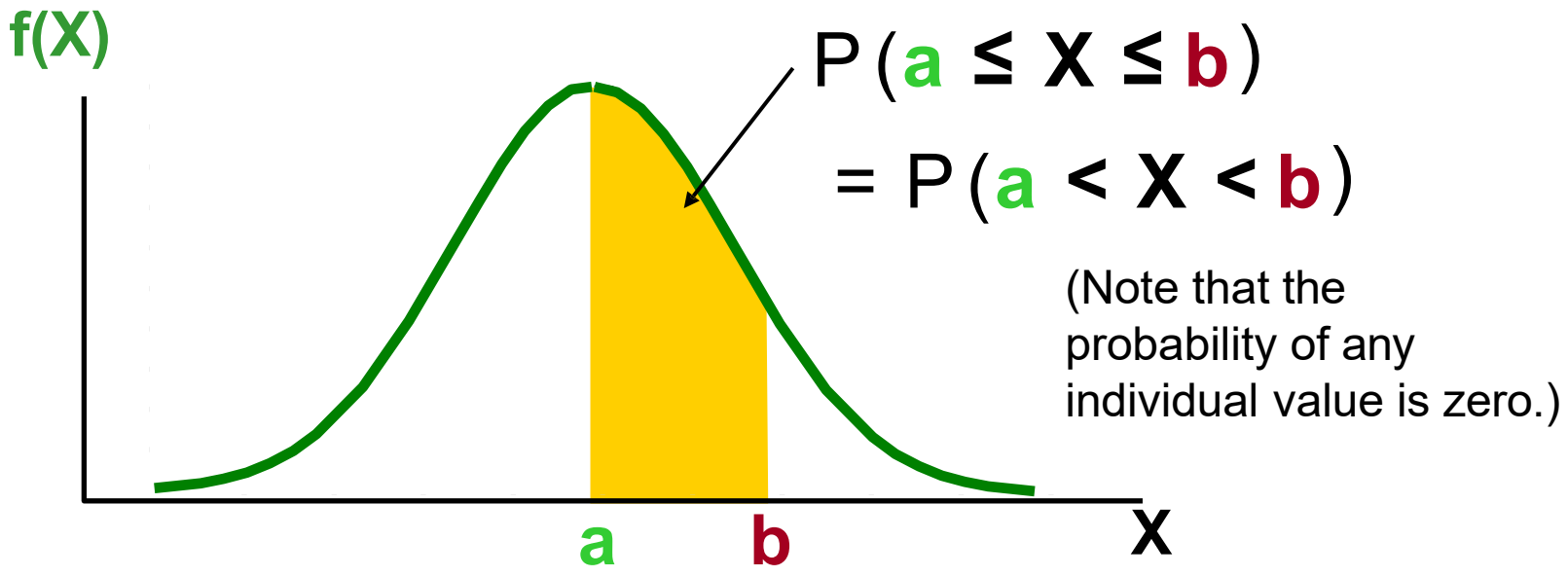
0 2.0 Z (μ = 0, σ = 1)

Note that the shape of the distribution is the same, only the scale has changed. We can express the problem in the original units (X in dollars) or in standardized units (Z).

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Finding Normal Probabilities

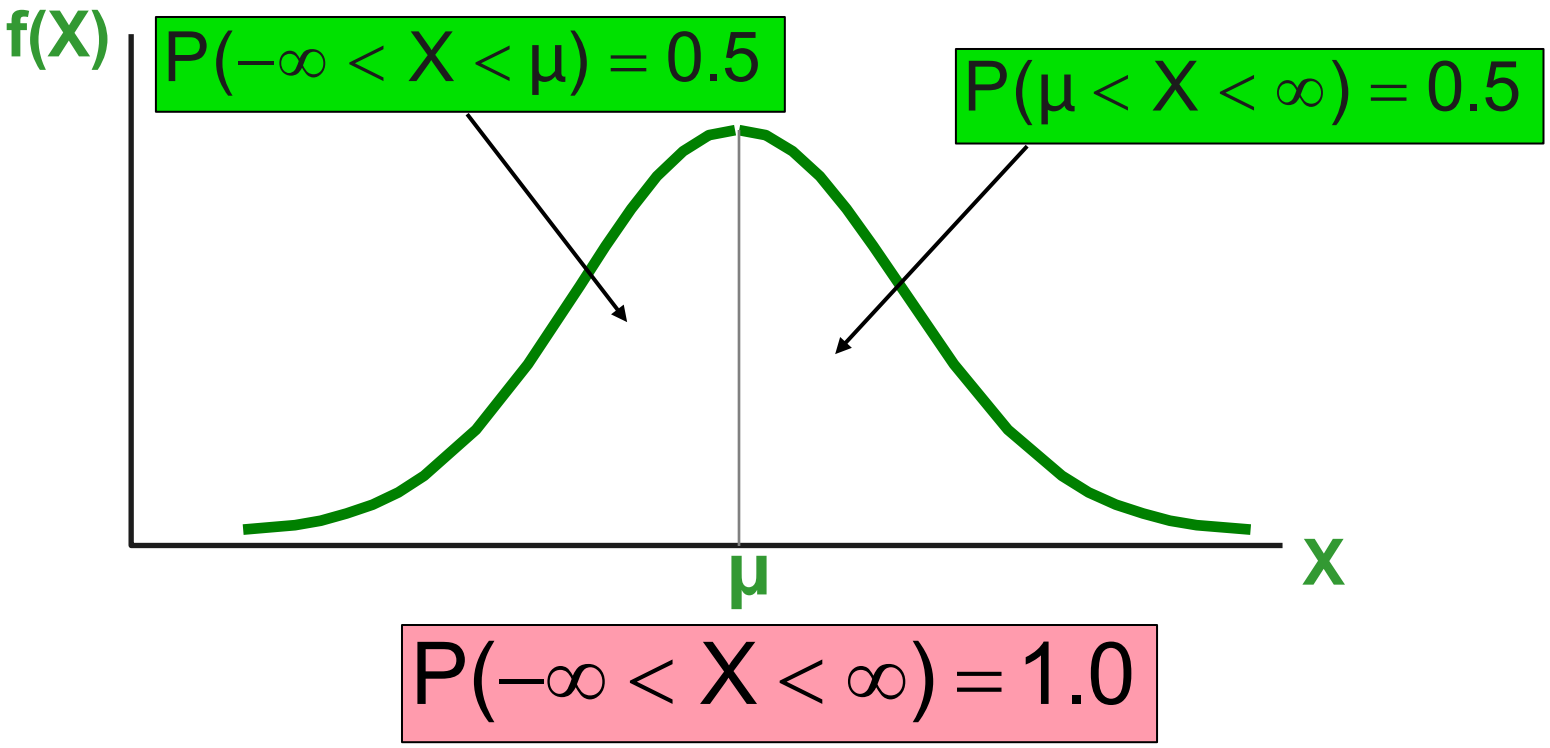
Probability is measured by the area under the curve.



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Probability as Area Under the Curve

The total area under the curve is 1.0, and the curve is symmetric, so half is above the mean, half is below.



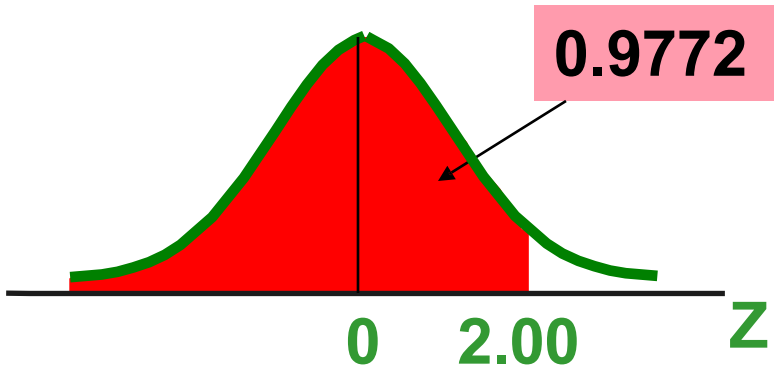
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The Standardized Normal Table

The Cumulative Standardized Normal table in the textbook (Appendix table E.2) gives the probability less than a desired value of Z (i.e., from negative infinity to Z).

Example:

P(Z < 2.00) = 0.9772

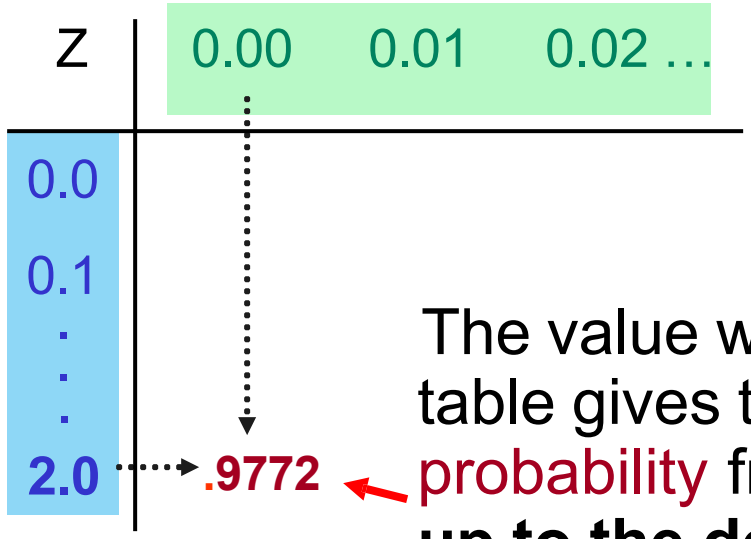


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The Standardized Normal Table

(continued)

The column gives the value of Z to the second decimal point.



The row shows the value of Z to the first decimal point.

The value within the table gives the probability from $Z = -\infty$ **up to the desired Z value. Left side of the area!**

P(Z < 2.00) = 0.9772

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General Procedure for Finding Normal Probabilities

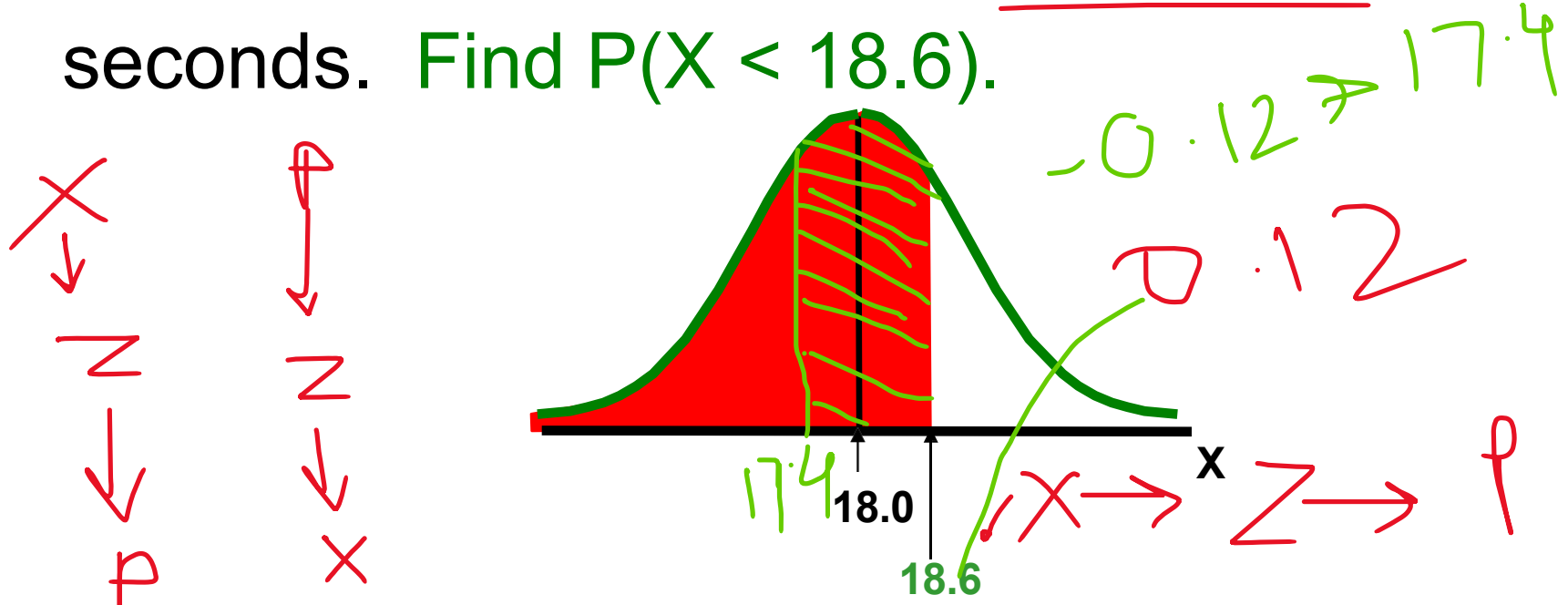
To find $P(a < X < b)$ when X is distributed normally:

- Draw the normal curve for the problem in terms of X.
- Translate X-values to Z-values.
- Use the Standardized Normal Table.

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Finding Normal Probabilities

- Let X represent the time it takes (in seconds) to download an image file from the internet.
- Suppose X is normal with a mean of 18.0 seconds and a standard deviation of 5.0 seconds. Find P(X < 18.6).



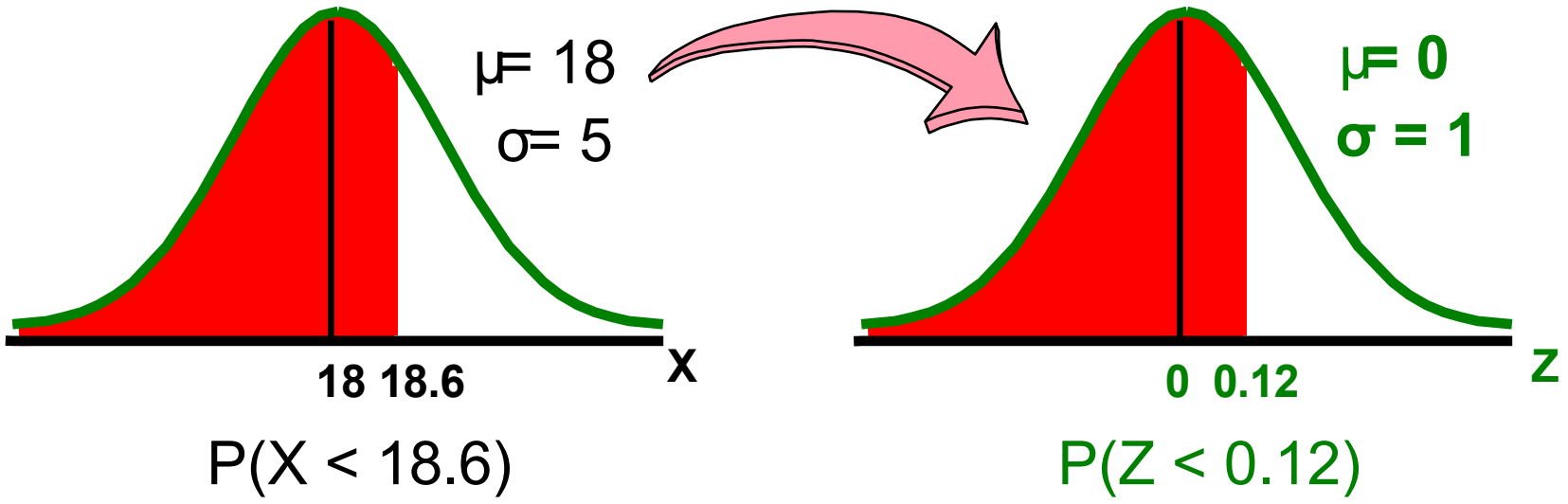
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Finding Normal Probabilities

(continued)

- Let X represent the time it takes, in seconds to download an image file from the internet.
- Suppose X is normal with a mean of 18.0 seconds and a standard deviation of 5.0 seconds. **Find P(X < 18.6):**

$$Z = \frac{X - \mu}{\sigma} = \frac{18.6 - 18.0}{5.0} = 0.12$$

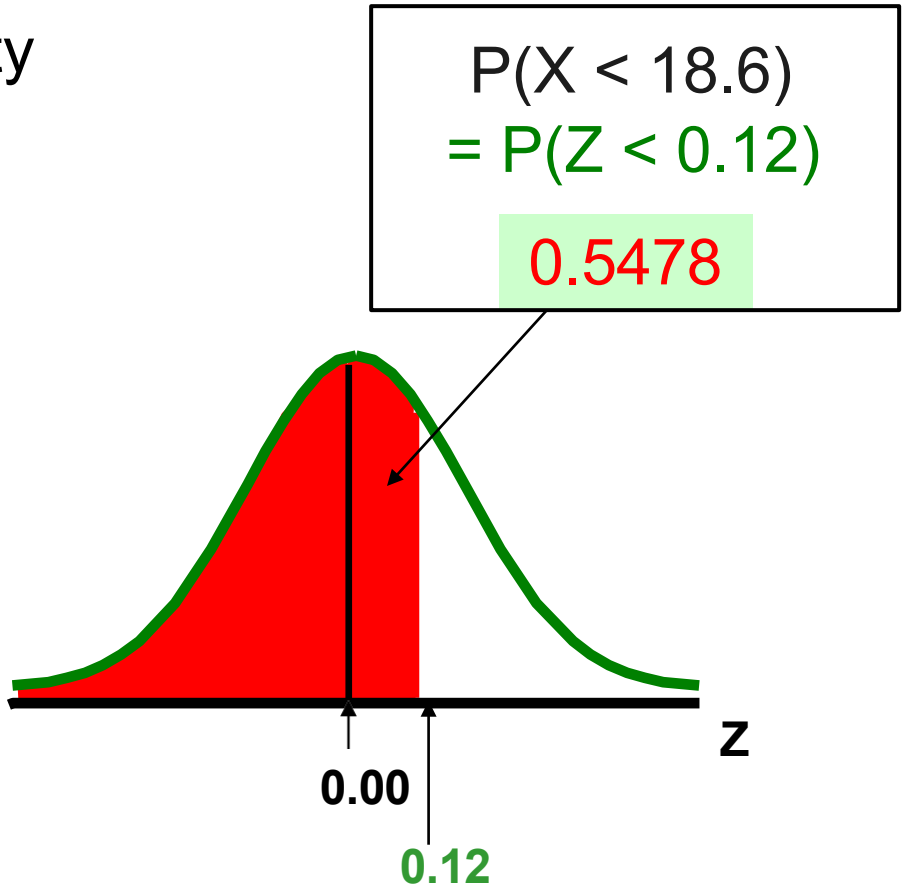


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Solution: Finding P(Z < 0.12)

Standardized Normal Probability Table (Portion)

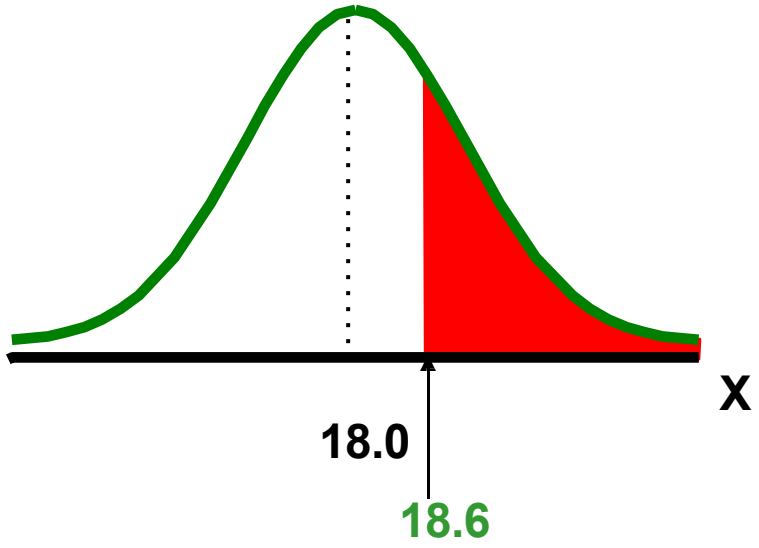
Z	.00	.01	.02
0.0	.5000	.5040	.5080
0.1	.5398	.5438	.5478
0.2	.5793	.5832	.5871
0.3	.6179	.6217	.6255



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Finding Normal Upper Tail Probabilities

- Suppose X is normal with mean 18.0 and standard deviation 5.0.
- Now Find P(X > 18.6).



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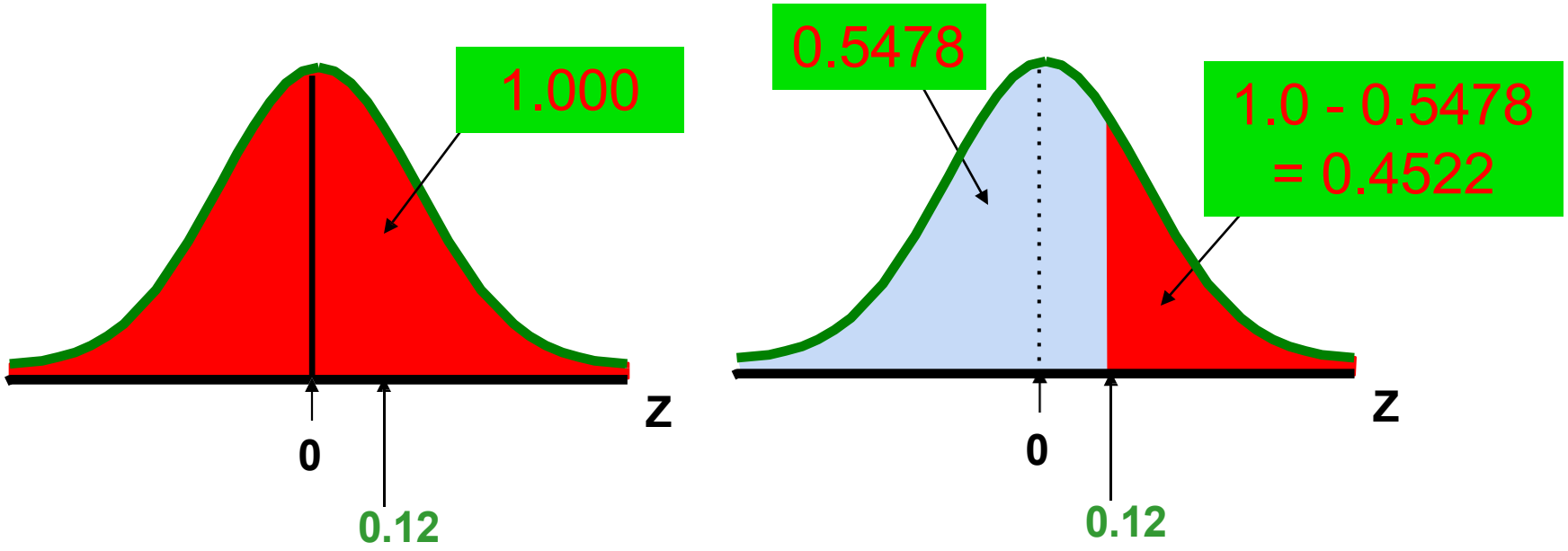
Finding Normal Upper Tail Probabilities

(continued)

- Now Find P(X > 18.6).

$$P(X > 18.6) = P(Z > 0.12) = 1.0 - P(Z \le 0.12)$$

$$= 1.0 - 0.5478 = 0.4522$$



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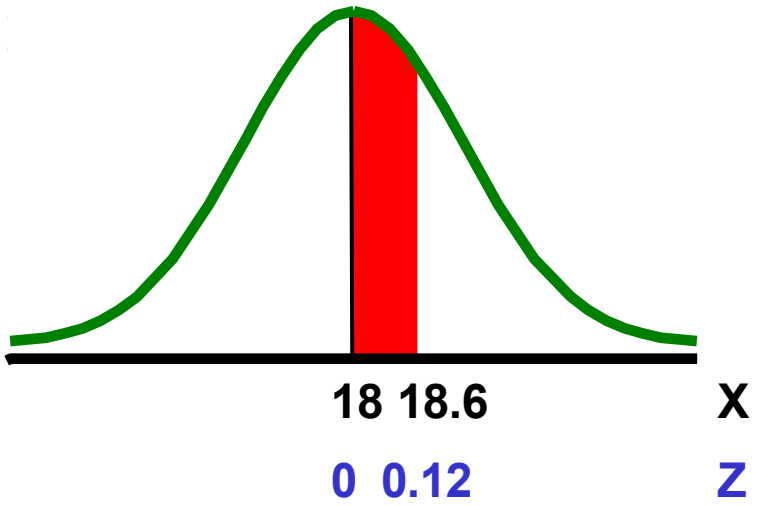
Finding a Normal Probability Between Two Values

- Suppose X is normal with mean 18.0 and standard deviation 5.0. Find P(18 < X < 18.6).

Calculate Z-values:

$$Z = \frac{X - \mu}{\sigma} = \frac{18 - 18}{5} = 0$$

$$Z = \frac{X - \mu}{\sigma} = \frac{18.6 - 18}{5} = 0.12$$



P(18 < X < 18.6)

= P(0 < Z < 0.12)

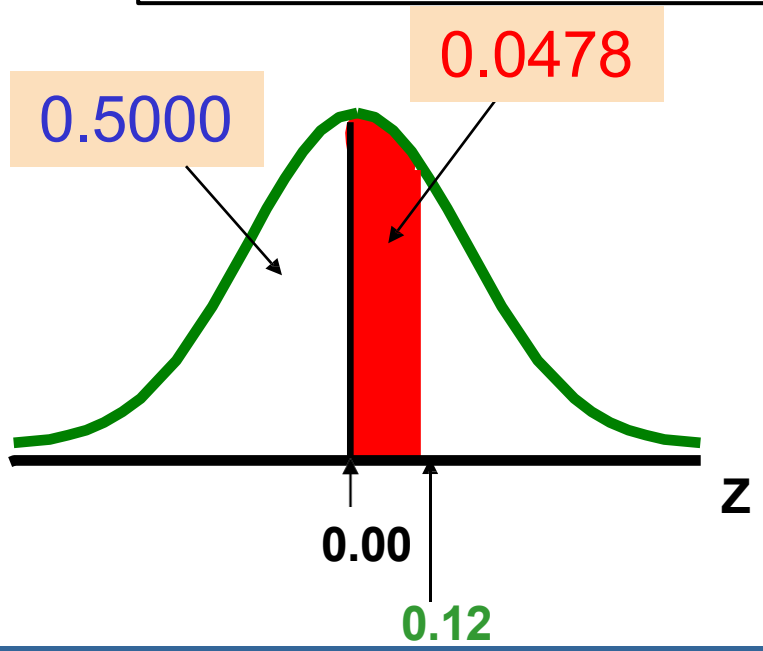
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Solution: Finding P(0 < Z < 0.12)

Standardized Normal Probability Table (Portion)

Z	.00	.01	.02
0.0	.5000	.5040	.5080
0.1	.5398	.5438	.5478
0.2	.5793	.5832	.5871
0.3	.6179	.6217	.6255

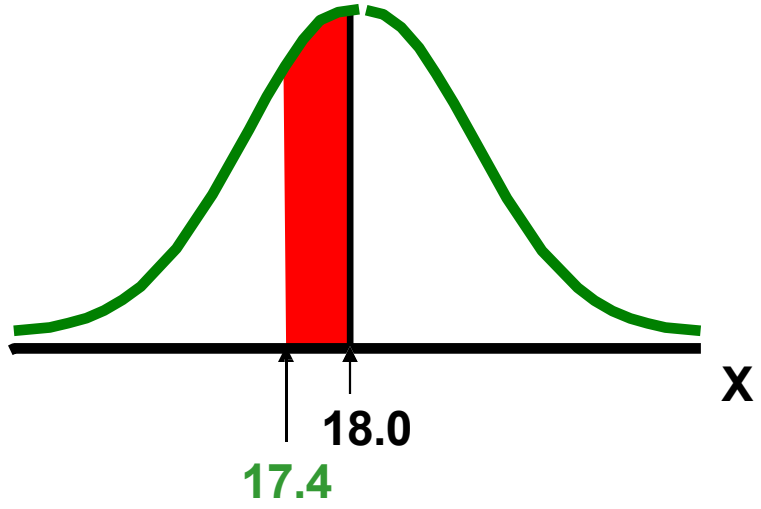
$P(18 < X < 18.6)$
$= P(0 < Z < 0.12)$
$= P(Z < 0.12) - P(Z \leq 0)$
$= 0.5478 - 0.5000 = 0.0478$



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Probabilities in the Lower Tail

- Suppose X is normal with mean 18.0 and standard deviation 5.0.
- Now Find P(17.4 < X < 18).



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Probabilities in the Lower Tail

(continued)

Now Find P(17.4 < X < 18):

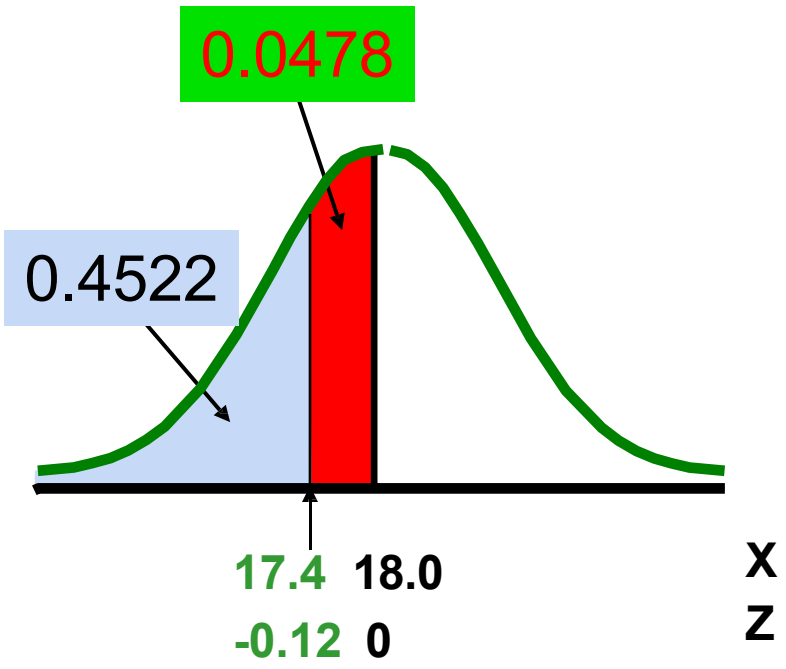
P(17.4 < X < 18)

= P(-0.12 < Z < 0)

= P(Z < 0) – P(Z ≤ -0.12)

= 0.5000 - 0.4522 = 0.0478

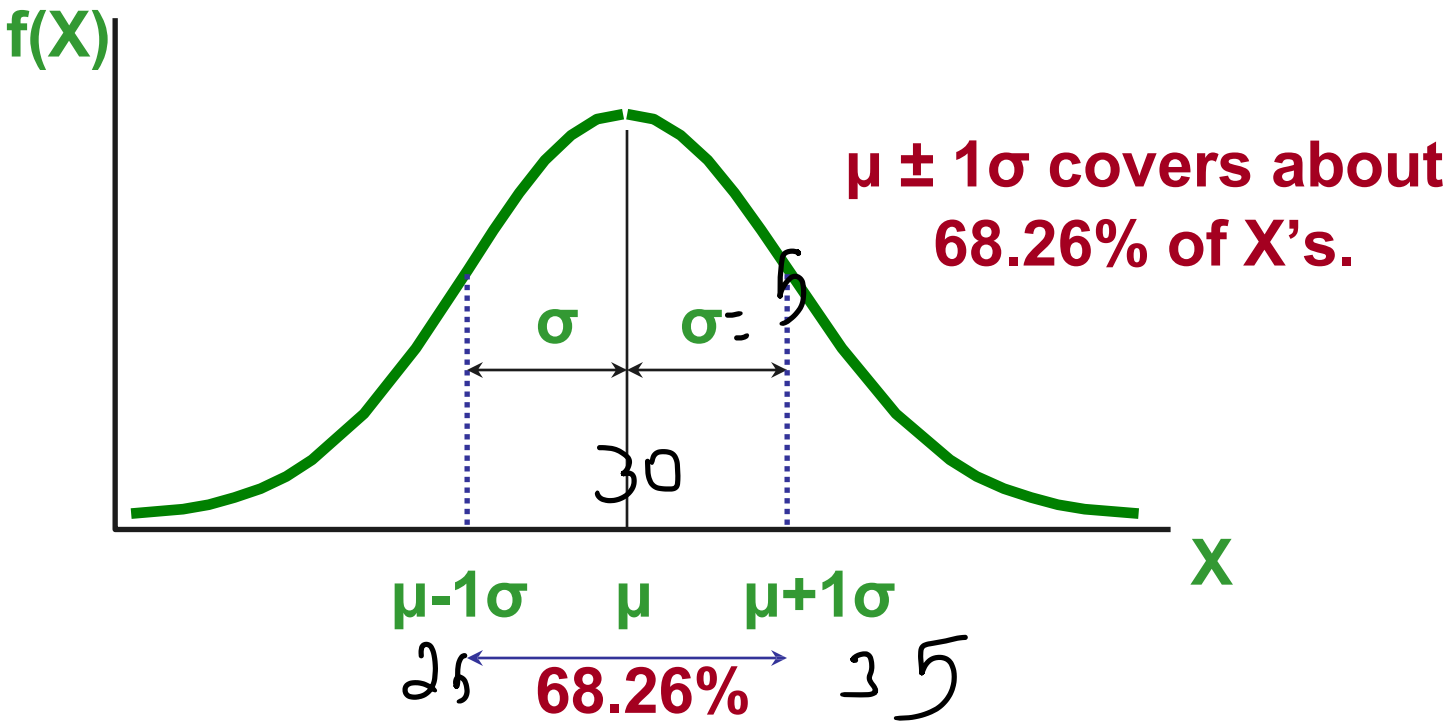
The Normal distribution is symmetric, so this probability is the same as P(0 < Z < 0.12).



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Empirical Rule

What can we say about the distribution of values around the mean? For any normal distribution:

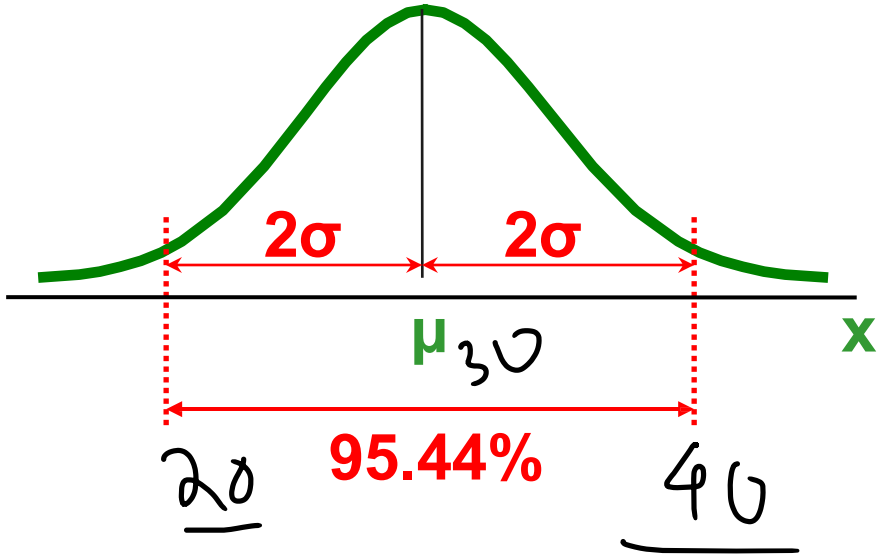


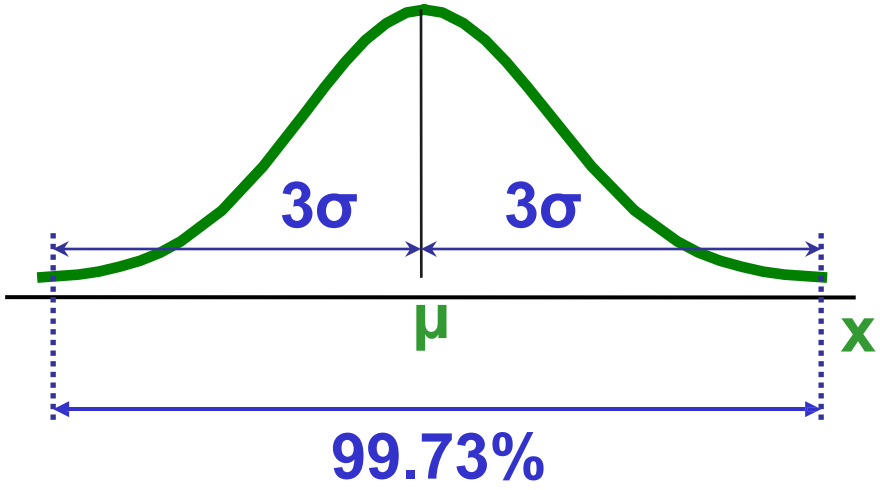
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The Empirical Rule

(continued)

- μ ± 2σ covers about 95.44% of X's.
- μ ± 3σ covers about 99.73% of X's.





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Given a Normal Probability Find the X Value

- Steps to find the X value for a known probability:
  1. Find the Z value for the known probability.
  2. Convert to X units using the formula:

$$X = \mu + Z\sigma$$

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Finding the X value for a Known Probability

(continued)

Example:

- Let X represent the time it takes (in seconds) to download an image file from the internet.
- Suppose X is normal with mean 18.0 and standard deviation 5.0.
- Find X such that 20% of download times are less than X.

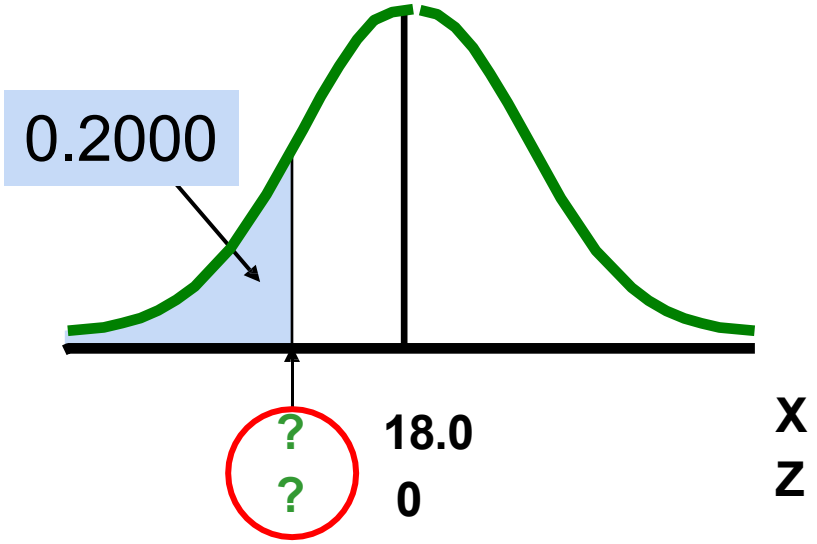
$P = 0.200$

$\downarrow$

Z

$\downarrow$

X



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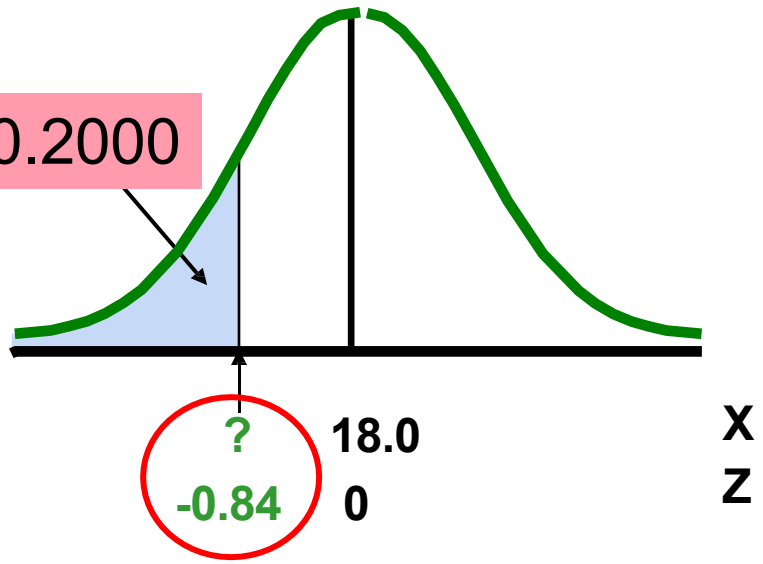
Find the Z value for 20% in the Lower Tail

1. Find the Z value for the known probability

Standardized Normal Probability Table (Portion)

Z	…	.03	.04	.05
-0.9	…	.1762	.1736	.1711
-0.8	…	.2033	.2005	.1977
-0.7	…	.2327	.2296	.2266

- 20% area in the lower tail is consistent with a Z value of -0.84.



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Finding the X value

2. Convert to X units using the formula:

$$X = \mu + Z\sigma$$
$$= 18.0 + (-0.84)5.0$$
$$= 13.8$$

So 20% of the values from a distribution with mean 18.0 and standard deviation 5.0 are less than 13.80.

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Using Excel –PHStat To Find Normal Probabilities

X is normal with a mean of 7 and a standard deviation of 2 find:

- P(X < 7)
- P(X > 9)
- P(X < 7 or X > 9)

	A	B	
1	Normal Probabilities		
2			
3	Common Data		
4	Mean	7	
5	Standard Deviation	2	
6			
7	Probability for X <=		
8	X Value	7	
9	Z Value	0	=STANDARDIZE(B8, B4, B5)
10	P(X<=7)	0.5000	=NORM.DIST(B8, B4, B5, TRUE)
11			
12	Probability for X >		
13	X Value	9	
14	Z Value	1	=STANDARDIZE(B13, B4, B5)
15	P(X>9)	0.1587	=1 - NORM.DIST(B13, B4, B5, TRUE)
16			
17	Probability for X<7 or X >9		
18	P(X<7 or X >9)	0.6587	=B10 + B15
19			

- P(5 < X < 9)
- X_L where $P(X < X_L) = 10\%$
- X_L & X_U where $P(X_L < X < X_U) = 95\%$

	A	B	
20	Probability for a Range		
21	From X Value	5	
22	To X Value	9	
23	Z Value for 5	-1	=STANDARDIZE(B21, B4, B5)
24	Z Value for 9	1	=STANDARDIZE(B22, B4, B5)
25	P(X<=5)	0.1587	=NORM.DIST(B21, B4, B5, TRUE)
26	P(X<=9)	0.8413	=NORM.DIST(B22, B4, B5, TRUE)
27	P(5<=X<=9)	0.6827	=ABS(B26 - B25)
28			
29	Find X and Z Given a Cum. Pctage.		
30	Cumulative Percentage	10.00%	
31	Z Value	-1.28	=NORM.S.INV(B30)
32	X Value	4.44	=NORM.INV(B30, B4, B5)
33			
34	Find X Values Given a Percentage		
35	Percentage	95.00%	
36	Z Value	-1.96	=NORM.S.INV((1 - B35)/2)
37	Lower X Value	3.08	=B4 + (B36 * B5)
38	Upper X Value	10.92	=B4 - (B36 * B5)

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Evaluating Normality

- Not all continuous distributions are normal.

- Normally distributed data should approximate the theoretical normal distribution:
  - The normal distribution is bell shaped (symmetrical) where the mean is equal to the median.
  - The empirical rule applies to the normal distribution.
  - The interquartile range of a normal distribution is 1.33 standard deviations.

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Evaluating Normality

(continued)

Comparing data characteristics to theoretical properties:

- Construct charts or graphs:
  - For small- or moderate-sized data sets, construct a stem-and-leaf display or a boxplot to check for symmetry.
  - For large data sets, does the histogram or polygon appear bell-shaped?
- Compute descriptive summary measures
  - Do the mean, median and mode have similar values?
  - Is the interquartile range approximately 1.33σ?
  - Is the range approximately 6σ?

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Evaluating Normality

(continued)

Comparing data characteristics to theoretical properties:

- Observe the distribution of the data set:
  - Do approximately 2/3 of the observations lie within mean ±1 standard deviation?
  - Do approximately 80% of the observations lie within mean ±1.28 standard deviations?
  - Do approximately 95% of the observations lie within mean ±2 standard deviations?
- Evaluate normal probability plot:
  - Is the normal probability plot approximately linear (i.e. a straight line) with positive slope?

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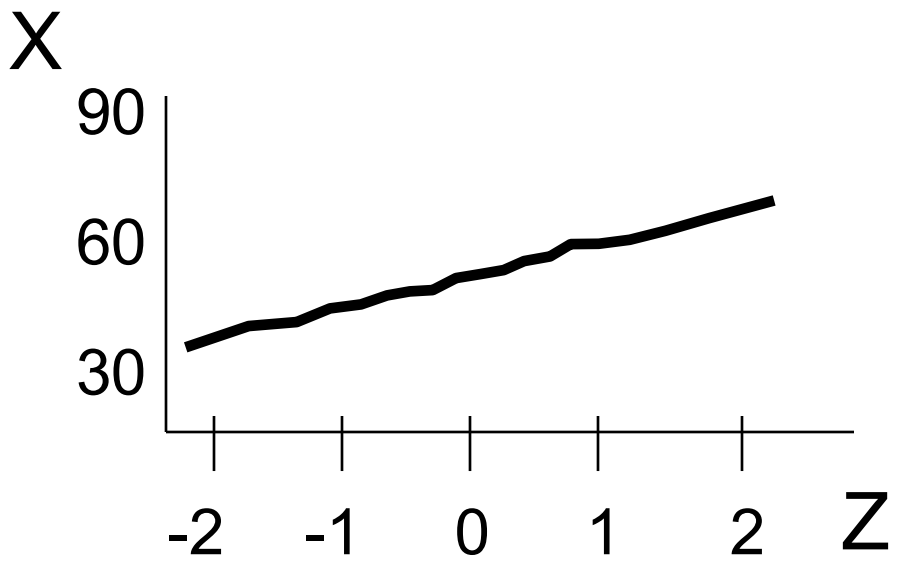
Constructing A Normal Probability Plot

- Normal probability plot:
  - Arrange data into ordered array.
  - Find corresponding standardized normal quantile values (Z).
  - Plot the pairs of points with observed data values (X) on the vertical axis and the standardized normal quantile values (Z) on the horizontal axis.
  - Evaluate the plot for evidence of linearity.

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The Normal Probability Plot Interpretation

A normal probability plot for data from a normal distribution will be approximately linear.



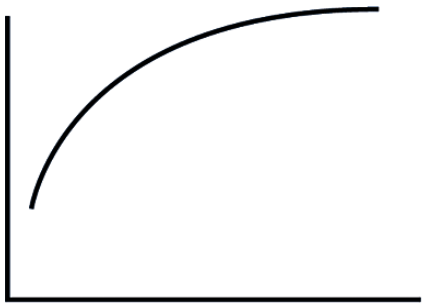
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Normal Probability Plot Interpretation

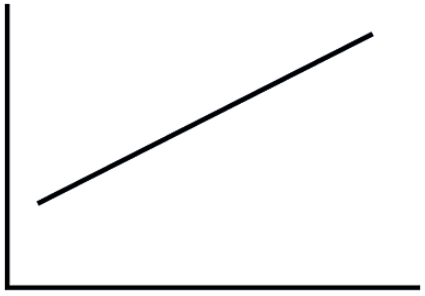
(continued)

Nonlinear plots indicate a deviation from normality.

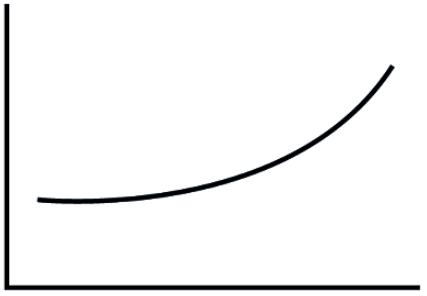
Left-Skewed



Normal



Right-Skewed

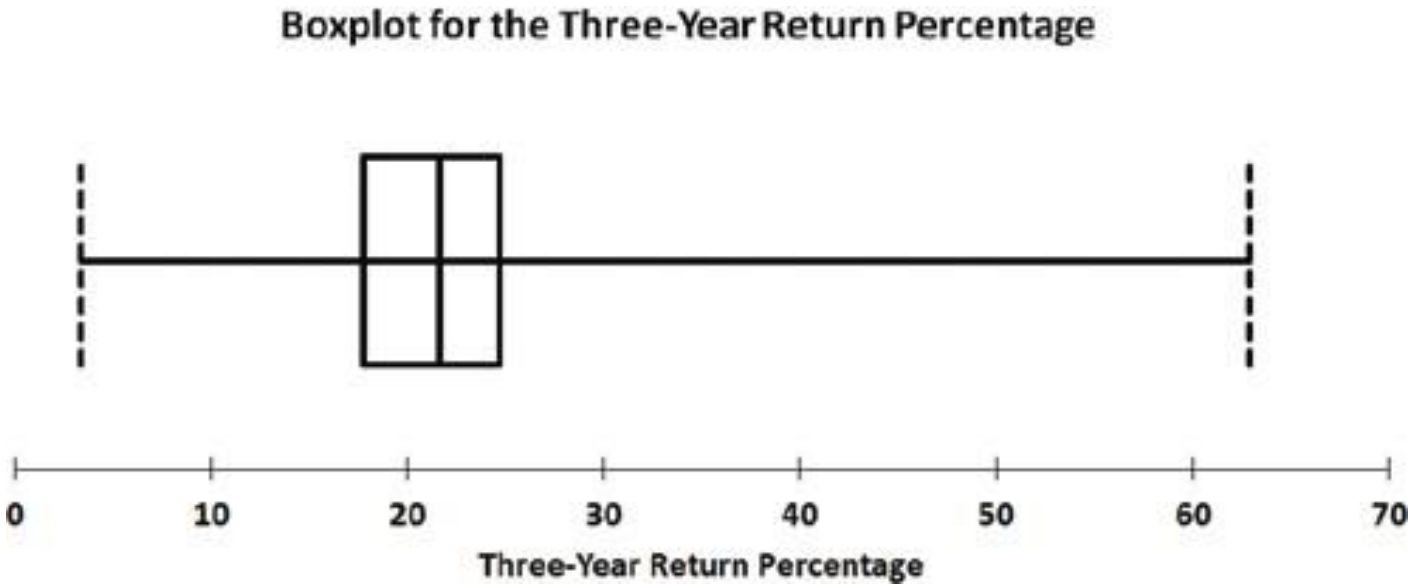


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Evaluating Normality
An Example: Mutual Fund Returns

Five-Number Summary	
Minimum	3.39
First quartile	17.76
Median	21.65
Third quartile	24.74
Maximum	62.91

The boxplot is skewed to the right. (The normal distribution is symmetric.)



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Evaluating Normality
An Example: Mutual Fund Returns

(continued)

Descriptive Statistics

	3YrReturn%
Mean	21.84
Median	21.65
Mode	21.74
Minimum	3.39
Maximum	62.91
Range	59.52
Variance	41.2968
Standard Deviation	6.4263
Coeff. of Variation	29.43%
Skewness	1.6976
Kurtosis	8.4670
Count	318
Standard Error	0.3604

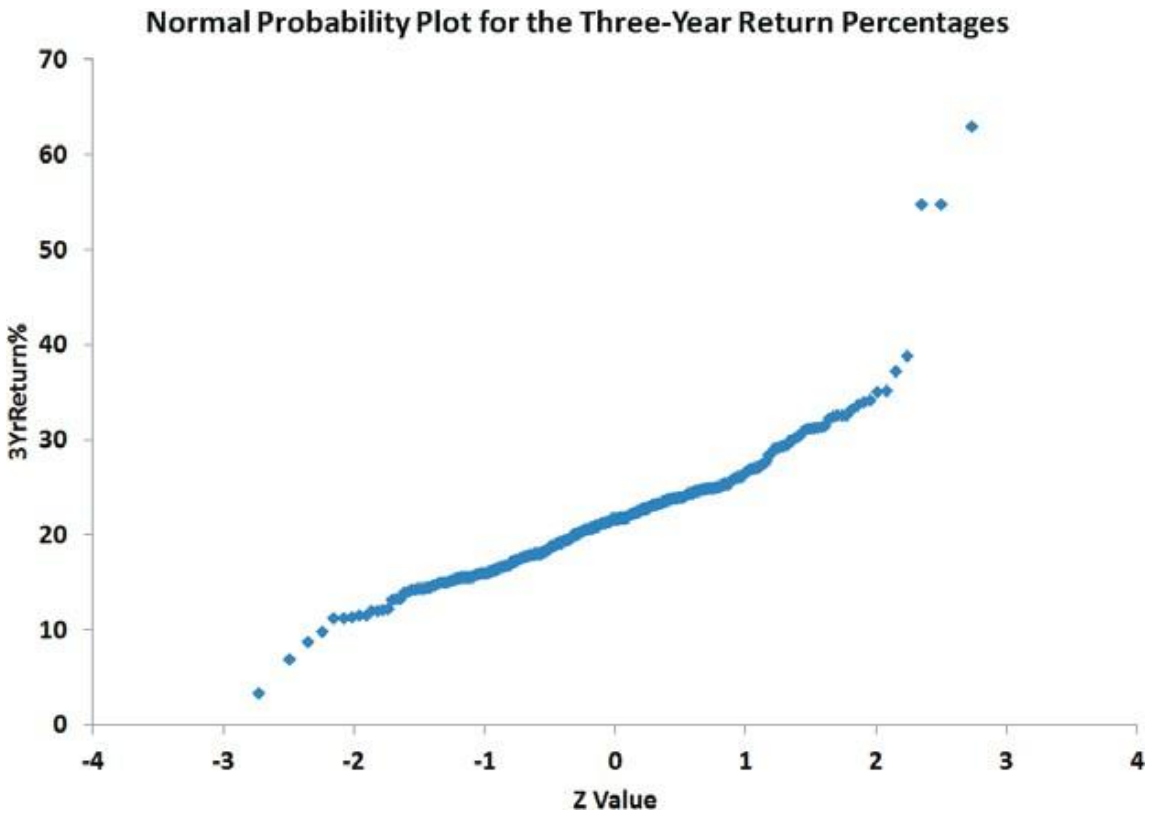
- The mean (21.84) is approximately the same as the median (21.65). (In a normal distribution the mean and median are equal.)
- The interquartile range of 6.98 is approximately 1.09 standard deviations. (In a normal distribution the interquartile range is 1.33 standard deviations.)
- The range of 59.52 is equal to 9.26 standard deviations. (In a normal distribution the range is 6 standard deviations.)
- 77.04% of the observations are within 1 standard deviation of the mean. (In a normal distribution this percentage is 68.26%.)
- 86.79% of the observations are within 1.28 standard deviations of the mean. (In a normal distribution this percentage is 80%.)
- 96.86% of the observations are within 2 standard deviations of the mean. (In a normal distribution this percentage is 95.44%.)
- The skewness statistic is 1.698 and the kurtosis statistic is 8.467. (In a normal distribution, each of these statistics equals zero.)

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Evaluating Normality Via Excel
An Example: Mutual Fund Returns

(continued)

Excel (quantile-quantile) normal probability plot



Plot is not a straight line and shows the distribution is skewed to the right. (A normal distribution appears as a straight line.)

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Evaluating Normality
An Example: Mutual Fund Returns

(continued)

- Conclusions
  - The returns are right-skewed.
  - The returns have more values concentrated around the mean than expected.
  - The range is larger than expected.
  - Normal probability plot is not a straight line.
  - Overall, this data set greatly differs from the theoretical properties of the normal distribution.

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The Uniform Distribution

- The **uniform distribution** is a probability distribution that has **equal probabilities** for all possible outcomes of the random variable.

- Also called a **rectangular distribution**.

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The Uniform Distribution

(continued)

The Continuous Uniform Distribution:

$$f(X) = \begin{cases} \dfrac{1}{b-a} & \text{if } a \le X \le b \\ 0 & \text{otherwise} \end{cases}$$

where

f(X) = value of the density function at any X value

a = minimum value of X

b = maximum value of X

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Properties of the Uniform Distribution

- The mean of a uniform distribution is:

$$\mu = \frac{a + b}{2}$$

- The standard deviation is:

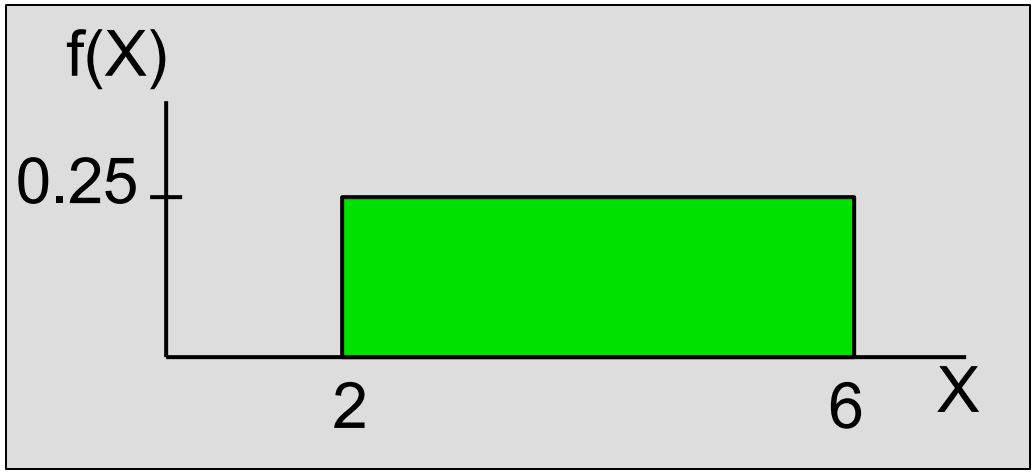
$$\sigma = \sqrt{\frac{(b - a)^2}{12}}$$

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Uniform Distribution Example

Example: Uniform probability distribution over the range $2 \le X \le 6$:

$$f(X) = \frac{1}{6 - 2} = 0.25 \quad \text{for} \quad 2 \le X \le 6$$



$$\mu = \frac{a + b}{2} = \frac{2 + 6}{2} = 4$$

$$\sigma = \sqrt{\frac{(b - a)^2}{12}} = \sqrt{\frac{(6 - 2)^2}{12}} = 1.1547$$

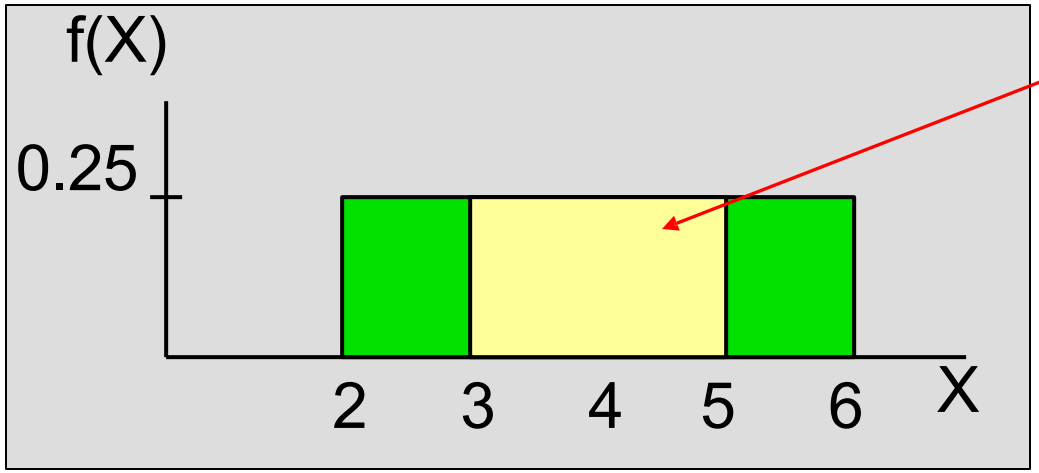
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Uniform Distribution Example

(continued)

Example: Using the uniform probability distribution to find $P(3 \le X \le 5)$:

$P(3 \le X \le 5) = (\text{Base})(\text{Height}) = (2)(0.25) = 0.5$



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Chapter Summary

In this chapter we discussed:

- Computing probabilities from the normal distribution.
- Using the normal distribution to solve business problems.
- Using the normal probability plot to determine whether a set of data is approximately normally distributed.
- Computing probabilities from the uniform distribution.

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$f(t) = -2e^{-t}$

$f'(t) = -2e^{t}$

$f''(t) = -2e^{t}$

e^{p} ← p

$-t$

$-2e^{-t}$

$\Rightarrow -2e^{p}$

chain rule → $f(x) = y$

$$\frac{dy}{dx} = \frac{dy}{dp} \cdot \frac{dp}{dx}$$

$$= 2e^{-t} \times (-1)$$

$$f(x) = e^x - x^3$$

$$f'(x) = e^x - 3x^2$$

$$f''(x) = e^x - 6x$$

$s - t_4 2\ 2t_3 1\ t_2 2\ t,$

$s(t) = t^4 - 2t^3 + t^2 - t$

$s'(t) =$ velocity

$s - t_4 2\ 2t_3 1\ t_2 2\ t, = 4t^3 - 6t^2 + 2t - 1$

$= s'(t)$

$s''(t) = 12t^2 - 12t + 2$

$s''(1) = 12 - 12 + 2 = 2\ m/s^2$

$$\frac{d}{dx}\left[f(x)\,g(x)\right]$$

$$\Rightarrow f'(x)\,g(x) + g'(x)\,f(x)$$

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)\,f'(x) - g'(x)\,f(x)}{(g(x))^2}$$

$$\frac{d}{dx}\left(e^x \cdot x^2\right) = e^x \cdot x^2 + 2x \cdot e^x$$

$$= e^x\left(x^2 + 2x\right)$$

$$\frac{d}{dx}\left(\frac{e^x}{x^2}\right) = \frac{x^2 \cdot e^x - e^x \cdot 2x}{x^4}$$

$$\Rightarrow \frac{x e^x \left[x - 2\right]}{x^4}$$

$$x^4$$

$\frac{d}{dx}(xe^x)$

$f'(x) = xe^x + e^x$

$f''(x) = xe^x + e^x + e^x$

$f'''(x) = xe^x + e^x + e^x + e^x$

$f^n(x) = xe^x + ne^x$

$y = \frac{e^x}{1+x^2}$ $\left(1, \frac{e}{2}\right)$

$y' = \frac{(1+x^2)e^x - e^x(2x)}{(1+x^2)^2}$

$(x-x_1)$

$(x-1) =$

57. Find $R'(0)$, where

$$R(x) = \frac{x - 3x^3 + 5x^5}{1 + 3x^3 + 6x^6 + 9x^9}$$

Hint: Instead of finding $R'(x)$ first, let $f(x)$ be the numerator and $g(x)$ be the denominator of $R(x)$, and compute $R'(0)$ from $f(0)$, $f'(0)$, $g(0)$, and $g'(0)$.

$\frac{e}{2}$

$\Rightarrow \frac{e^x(x-1)^2}{(1+x^2)^2}$ $(1+x^2)$

$y = mx + c$

57. Find $R'(0)$, where

$$R(x) = \frac{x - 3x^3 + 5x^5}{1 + 3x^3 + 6x^6 + 9x^9}$$

Hint: Instead of finding $R'(x)$ first, let $f(x)$ be the numerator and $g(x)$ be the denominator of $R(x)$, and compute $R'(0)$ from $f(0)$, $f'(0)$, $g(0)$, and $g'(0)$.

$R(x) = \frac{f(x)}{g(x)} = \frac{g(x)f'(x) - g'(x)f(x)}{(g(x))^2}$

$g(0) = 1$

$f(0) = 0$

$f'(x) = 1 - 9x^2 + 25x^4$

$f'(0) = 1 \quad = \frac{1 \times 1 - 0}{1^2} = 1$

62. The *biomass* $B(t)$ of a fish population is the total mass of the members of the population at time t. It is the product of the number of individuals $N(t)$ in the population and the average mass $M(t)$ of a fish at time t. In the case of guppies, breeding occurs continually. Suppose that at time $t = 4$ weeks the population is 820 guppies and is growing at a rate of 50 guppies per week, while the average mass is 1.2 g and is increasing at a rate of 0.14 g/week. At what rate is the biomass increasing when $t = 4$?

$B(t) = N(t)M(t)$

$B(4) =$ biomass

$N(t) \rightarrow$ population

$M(t) \rightarrow$ av. mass

$N(4) = 820$, $N'(4) = 50$

$M(4) = 1.2\text{g}$ $M'(4) = 0.14$

$$B(t) = N(t)\,M(t)$$

$$B'(t) = N'(t)\,M(t) + M'(t)\,N(t)$$